

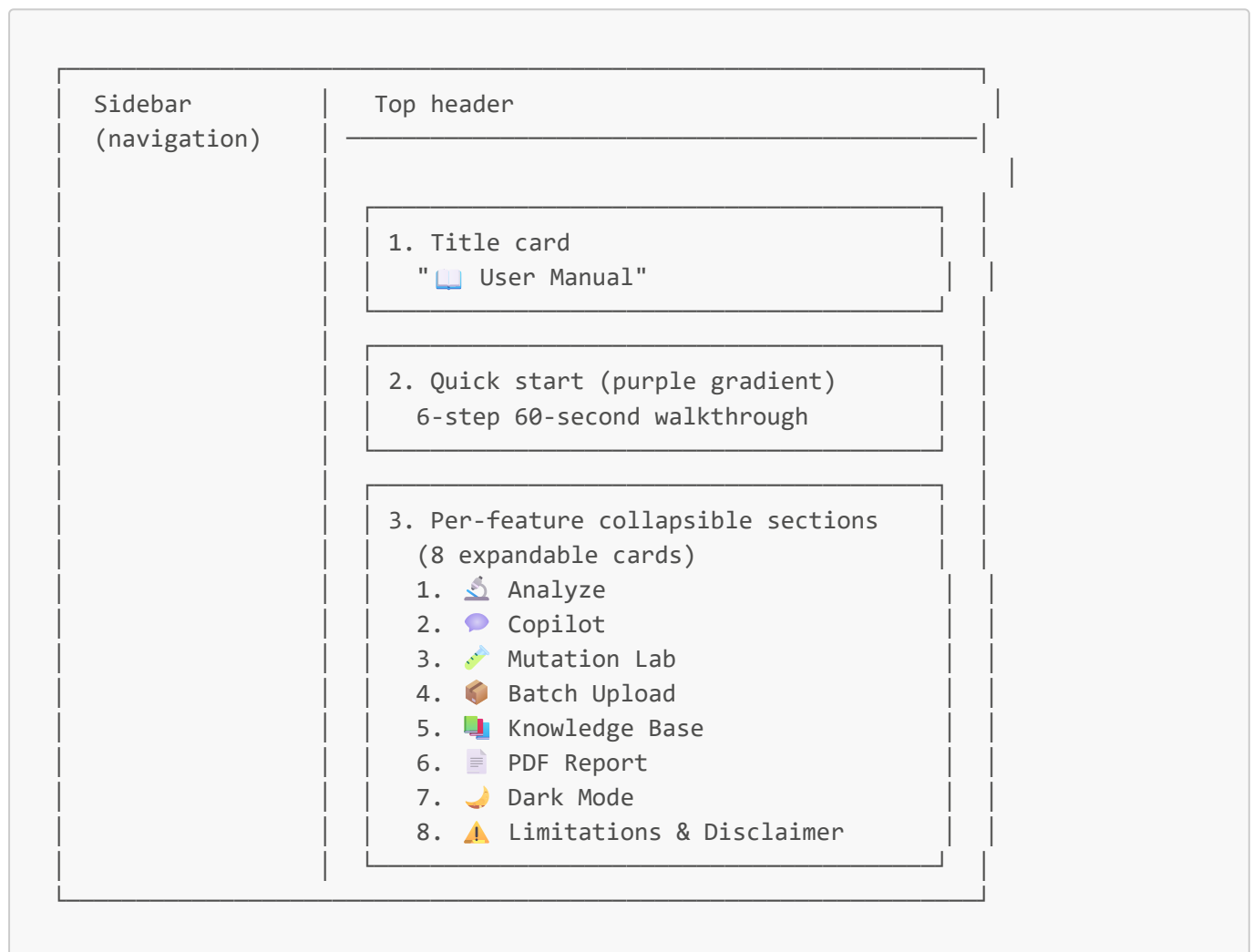
# User Manual Page — Complete Component Guide

A meta-deep-dive into the **User Manual** page of GenomelQ. This is the help page that lives inside the app itself. This document covers what's on it, how it's organized, and how it complements the external docs.

For each component this document covers:

-  **What it is** (plain-language description)
-  **What it tells you** (how to read the output)
-  **How it works under the hood** (the algorithm)
-  **How to use it** (the user flow)
-  **Practical tips** (when it's useful, when to be cautious)

## Page Layout Overview



The User Manual page is structured as a **single-column, max-4xl-wide** scrollable column with three vertical regions: the title, the gradient quick-start card, and a stack of collapsible feature sections.

## 1. Title Card

## What it is

A simple white card at the top of the page with the heading " User Manual" and a single-line description.

## What it tells you

This is your **landing page** for in-app help — it announces what this page does (a walkthrough of every feature in GenomelQ) and invites you to click sidebar items to jump elsewhere.

## How it works under the hood

Static HTML. No JavaScript or backend involvement.

## Practical tips

- This page is meant to be **read top-to-bottom** the first time, then **referenced section-by-section** afterward.
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## 2. Quick Start Card (Purple Gradient)

### What it is

A highlighted card with a purple-blue gradient background that compresses GenomelQ into a **6-step, 60-second tutorial**.

### What it tells you

The 6 steps:

1. Open the **Analyze** tab.
2. Click any sample button (BRCA1 / INS / APP / ACTB) or paste your own DNA sequence.
3. Press **Analyze Sequence**.
4. Read the verdict, probabilities, saliency map and circular plot.
5. Click **PDF** in the top-right to download a report.
6. Switch to the **Copilot** tab and ask a follow-up question (e.g. "Why is this Cancer?").

This is designed to be the **shortest path to seeing the project work end-to-end**.

### How it works under the hood

Static HTML, styled with Tailwind's `bg-gradient-to-br` from `from-brand-50` to `to-purple-50` (light) or `from-brand-500/10` to `to-purple-500/10` (dark).

### Practical tips

- When **demoing the project to others** (especially during a defense), this is the script you can use as your spoken intro.
  - Anyone arriving at the app for the first time can **skip the rest of this manual** if they just follow these 6 steps.
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### 3. Collapsible Feature Sections (8 Cards)

#### What it is

Eight independent collapsible (<details> / <summary>) cards. Each covers one major feature of GenomelQ and can be expanded or collapsed independently.

#### What it tells you

The eight sections (and what's in each):

#### 3.1 **Analyze — predict disease from a DNA sequence (open by default)**

Detailed walkthrough of the Analyze page:

- What it does
- How to use it (input sequence, choose backbone, toggle options)
- What the result looks like (verdict, probabilities, saliency, ORFs, similar hits)
- Pro tips (low confidence, OOD high risk)

#### 3.2 **Copilot — ask AI to explain anything**

Walkthrough of the Copilot:

- Three providers (DEMO, GEMINI, OLLAMA)
- How to use suggested questions
- How prediction context is integrated
- Sample good questions to try
- Refusal of medical-advice queries

#### 3.3 **Mutation Lab — compare reference vs variant**

Walkthrough of mutation analysis:

- Reference and variant boxes
- The compare button
- Result interpretation (mutation events table, probability shifts, AA effects)
- When to use this feature

#### 3.4 **Batch Upload — classify many sequences at once**

Walkthrough of batch FASTA upload:

- File format requirements
- Accepted extensions (.fa / .fasta / .txt)
- Result columns explained

#### 3.5 **Knowledge Base — disease & gene reference**

Walkthrough of the Knowledge Base page:

- Static reference cards
- Browsing tips

### 3.6 📄 PDF Report download

Walkthrough of the PDF export:

- Where the button is
- What's in the PDF (verdict, OOD, probabilities, stats, similar hits, ORFs, top saliency regions, sequence)

### 3.7 🌙 Dark mode + theme toggle

Walkthrough of theme switching:

- Where the toggle is
- That it persists in localStorage
- Charts/plot adapt

### 3.8 ⚠️ Limitations & disclaimer

A frank list of what GenomelQ **doesn't** do:

- Trained on a limited corpus (~22k augmented samples)
- 4 broad classes only
- OOD-risk inputs may give confident-looking but wrong answers
- Copilot LLM may hallucinate despite retrieval grounding
- **Not for clinical use**

### ⚙️ How it works under the hood

Each section is a native HTML `<details>` element:

```
<details class="...">
  <summary class="...">
    <span class="text-xl">📄</span> 1. Analyze – predict disease from a DNA
sequence
  </summary>
  <div class="mt-3 ...">
    <p><strong>What it does:</strong> ...</p>
    <ol class="list-decimal list-inside ...">
      <li>...</li>
    </ol>
  </div>
</details>
```

Native HTML provides the toggle behaviour — no JavaScript needed. Tailwind handles the styling. The first one is `open` by default; the rest start collapsed.

### 📖 How to use it

1. Read the **Quick Start** card to get oriented.
2. **Click** any feature heading to expand it for details.
3. **Click again** to collapse and reduce visual clutter.
4. Several can be open at the same time.

### 💡 Practical tips

- This page is **fully self-contained** — it doesn't make any API calls. You can read it offline.
- The information here intentionally **mirrors** the per-page guides in the [Guide/](#) folder. Use this in-app manual for quick reminders, and the per-page Markdown / PDF guides for deep-dive reference (e.g., during a defense or when writing your thesis).

## 🔄 The User Manual Flow (End-to-End)

```
User clicks "📖 User Manual" in sidebar
      ↓
Alpine sets page = 'manual'
      ↓
Section becomes visible (x-show toggles)
      ↓
Static HTML renders instantly (no API calls)
      ↓
User scrolls / expands sections at will
```

Total time: **instant** rendering; no network calls.

## 💡 Cheat Sheet — When to Use the In-App Manual vs. External Docs

| Need                                        | Best resource                                      |
|---------------------------------------------|----------------------------------------------------|
| Quick reminder while using the app          | <b>In-app User Manual</b> (this page)              |
| First-time user, no idea where to start     | <b>In-app Quick Start card</b> (top of this page)  |
| Deep technical reference (algorithms, code) | <a href="#">DOCS.md</a> at project root            |
| Per-page detailed walkthrough               | <a href="#">Guide/&lt;page&gt;_GUIDE.pdf</a> files |
| Setup VS Code, install dependencies         | <a href="#">HOW_TO_RUN_IN_VSCODE.md</a>            |
| Test every function                         | <a href="#">TESTING_GUIDE.md</a>                   |
| Build the GitHub README                     | <a href="#">README.md</a>                          |

## 📖 Related Documents

- [README.md](#) — high-level project overview
- [DOCS.md](#) — comprehensive technical documentation

- [HOW\\_TO\\_RUN\\_IN\\_VSCODE.md](#) — VS Code setup and debugging
  - [TESTING\\_GUIDE.md](#) — full test plan
  - [Guide/](#) — per-page deep-dive markdown / PDF guides
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*Part of the GenomelQ per-page documentation series.*